



TRS-80 MC-10 — Chip Pinouts

MAJOR ICS · 6803 · 6847 VDG · MEMORY · TEST & REPAIR

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an underline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

MC6803 CPU (ROMless 6801) 40-DIP

Vss	1	40	E
XTAL1	2	39	SC1
EXTAL2	3	38	SC2
/NMI	4	37	P30
/IRQ1	5	36	P31
/RESET	6	35	P32
Vcc	7	34	P33
P20	8	33	P34
P21	9	32	P35
P22	10	31	P36
P23	11	30	P37
P24	12	29	P40
P10	13	28	P41
P11	14	27	P42
P12	15	26	P43
P13	16	25	P44
P14	17	24	P45
P15	18	23	P46
P16	19	22	P47
P17	20	21	Vcc Stby

MC-10 CPU. The 6803 is a ROM-less 6801 (pin-compatible 40-pin). Multiplexed address/data on Port 3/4. **Fails:** dead machine, no boot.

MC6847 VDG video 40-DIP

Vss	1	40	DD7
DD6	2	39	CSS
DD0	3	38	HS
DD1	4	37	FS
DD2	5	36	RP
DD3	6	35	A/G
DD4	7	34	A/S
DD5	8	33	CLK
CHB	9	32	INV
φB	10	31	INT/EXT
φA	11	30	GM0
/MS	12	29	GM1
DA5	13	28	Y
DA6	14	27	GM2
DA7	15	26	DA4
DA8	16	25	DA3
Vcc	17	24	DA2
DA9	18	23	DA1
DA10	19	22	DA0
DA11	20	21	DA12

Video Display Generator: 32×16 text / semigraphics, composite via an MC1372. Pinout verified against the Motorola MC6847/MC6847Y datasheet. **Fails:** no/garbled picture.

Memory & ROM RAM/ROM

RAM	4KB static (expandable to 20KB)
VRAM	shared with 6847
ROM	8KB Microcolor BASIC (2364-class)
Video	6847 + MC1372 modulator

MC-10 RAM is small static RAM, not DRAM. ROM is a single mask ROM.

Test & Repair tips

No boot	6803 / clock / ROM
No / bad video	6847 (+ MC1372)
Bad keys	keyboard matrix → 6803 port
Garbled chars	6847 or its VRAM

The 6847 and MC1372 are the usual video-fault suspects; the 6803 is the usual dead-machine suspect.

Proofread checklist △ verify these

MC6803
ROM / RAM type
(6803 & 6847 pinouts now datasheet-verified)
Amber pins and △ flagged chips are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.